MSG No. 187.210 $P\bar{6}m21'$ [Type II, hexagonal]

Table 1: Wyckoff site: `1a`, site symmetry: `-6m21'`

No.	position	mapping
1	[0, 0, 0]	[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24]

Table 2: Wyckoff site: `1b`, site symmetry: `-6m21'`

No.	position	mapping
1	$\left[0, 0, \frac{1}{2}\right]$	[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24]

Table 3: Wyckoff site: `1c`, site symmetry: `-6m21'`

No.	position	mapping
1	$\left[\frac{1}{3}, \frac{2}{3}, 0\right]$	[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24]

Table 4: Wyckoff site: `1d`, site symmetry: `-6m21'`

No.	position	mapping
1	$\left[\frac{1}{3}, \frac{2}{3}, \frac{1}{2}\right]$	[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24]

Table 5: Wyckoff site: `1e`, site symmetry: `-6m21'`

No.	position	mapping
1	$\left[\frac{2}{3}, \frac{1}{3}, 0\right]$	[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24]

Table 6: Wyckoff site: `1f`, site symmetry: `-6m21'`

No.	position	mapping
1	$\left[\frac{2}{3}, \frac{1}{3}, \frac{1}{2}\right]$	[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24]

Table 7: Wyckoff site: `2g`, site symmetry: `3m.1'`

No.	position	mapping
1	$[0, 0, z]$	[1,2,3,10,11,12,13,14,15,22,23,24]
2	$[0, 0, -z]$	[4,5,6,7,8,9,16,17,18,19,20,21]

Table 8: Wyckoff site: 2h, site symmetry: 3m.1'

No.	position	mapping
1	$\left[\frac{1}{3},\ \frac{2}{3},\ z\right]$	[1,2,3,10,11,12,13,14,15,22,23,24]
2	$\left[\frac{1}{3},\ \frac{2}{3},\ -z\right]$	[4,5,6,7,8,9,16,17,18,19,20,21]

Table 9: Wyckoff site: 2i, site symmetry: 3m.1'

No.	position	mapping
1	$\left[\frac{2}{3},\ \frac{1}{3},\ z\right]$	[1,2,3,10,11,12,13,14,15,22,23,24]
2	$\left[\frac{2}{3},\ \frac{1}{3},\ -z\right]$	[4,5,6,7,8,9,16,17,18,19,20,21]

Table 10: Wyckoff site: 3j, site symmetry: mm21'

No.	position	mapping
1	$[x,\ -x,\ 0]$	[1,6,8,11,13,18,20,23]
2	$[x,\ 2x,\ 0]$	[2,4,9,12,14,16,21,24]
3	$[-2x,\ -x,\ 0]$	[3,5,7,10,15,17,19,22]

Table 11: Wyckoff site: 3k, site symmetry: mm21'

No.	position	mapping
1	$\left[x,\ -x,\ \frac{1}{2}\right]$	[1,6,8,11,13,18,20,23]
2	$\left[x,\ 2x,\ \frac{1}{2}\right]$	[2,4,9,12,14,16,21,24]
3	$\left[-2x,\ -x,\ \frac{1}{2}\right]$	[3,5,7,10,15,17,19,22]

Table 12: Wyckoff site: 6l, site symmetry: m..1'

No.	position	mapping
1	$[x,\ y,\ 0]$	[1,8,13,20]
2	$[-y,\ x-y,\ 0]$	[2,9,14,21]
3	$[-x+y,\ -x,\ 0]$	[3,7,15,19]
4	$[x,\ x-y,\ 0]$	[4,12,16,24]
5	$[-x+y,\ y,\ 0]$	[5,10,17,22]
6	$[-y,\ -x,\ 0]$	[6,11,18,23]

Table 13: Wyckoff site: 6m, site symmetry: m..1'

No.	position	mapping
1	$\left[x,\ y,\ \frac{1}{2}\right]$	[1,8,13,20]
2	$\left[-y,\ x-y,\ \frac{1}{2}\right]$	[2,9,14,21]
3	$\left[-x+y,\ -x,\ \frac{1}{2}\right]$	[3,7,15,19]
4	$\left[x,\ x-y,\ \frac{1}{2}\right]$	[4,12,16,24]
5	$\left[-x+y,\ y,\ \frac{1}{2}\right]$	[5,10,17,22]

continued ...

Table 13

No.	position	mapping
6	$\left[-y,\ -x,\ \frac{1}{2}\right]$	`[6,11,18,23]`

Table 14: Wyckoff site: `6n`, site symmetry: `.m.1'`

No.	position	mapping
1	$[x,\ -x,\ z]$	`[1,11,13,23]`
2	$[x,\ 2x,\ z]$	`[2,12,14,24]`
3	$[-2x,\ -x,\ z]$	`[3,10,15,22]`
4	$[x,\ 2x,\ -z]$	`[4,9,16,21]`
5	$[-2x,\ -x,\ -z]$	`[5,7,17,19]`
6	$[x,\ -x,\ -z]$	`[6,8,18,20]`

Table 15: Wyckoff site: `12o`, site symmetry: `11'`

No.	position	mapping
1	$[x,\ y,\ z]$	`[1,13]`
2	$[-y,\ x-y,\ z]$	`[2,14]`
3	$[-x+y,\ -x,\ z]$	`[3,15]`
4	$[x,\ x-y,\ -z]$	`[4,16]`
5	$[-x+y,\ y,\ -z]$	`[5,17]`
6	$[-y,\ -x,\ -z]$	`[6,18]`
7	$[-x+y,\ -x,\ -z]$	`[7,19]`
8	$[x,\ y,\ -z]$	`[8,20]`
9	$[-y,\ x-y,\ -z]$	`[9,21]`
10	$[-x+y,\ y,\ z]$	`[10,22]`
11	$[-y,\ -x,\ z]$	`[11,23]`
12	$[x,\ x-y,\ z]$	`[12,24]`